

BPHO ROUND 0 · MARKSCHEME

Drill bank section 1

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.*

Quick key

Q1 B	Q2 C	Q3 B	Q4 E	Q5 D
Q6 B	Q7 E	Q8 C	Q9 A	Q10 D
Q11 C	Q12 D	Q13 E	Q14 A	Q15 B
Q16 E	Q17 D	Q18 C	Q19 B	Q20 A
Q21 B	Q22 E	Q23 C	Q24 A	Q25 D

Worked solutions

Q1 · Atwood machine: the tension in a string that is not in equilibrium — **Answer B**

What is being tested. Whether you realise that a pulley system in motion is *not* in equilibrium. The single most common answer to this question is **mg**, and it is wrong for a reason worth understanding.

Step 1 — one acceleration, one tension. The string is inextensible, so if the heavier block descends a distance **d** the lighter block rises by the same **d**. Both therefore have the same magnitude of acceleration **a**, in opposite directions. The string is light, so it has no mass to accelerate, and the pulley is smooth, so it exerts no frictional torque; therefore the tension **T** is the same on both sides. Two unknowns, **a** and **T**.

Step 2 — Newton's second law for each block. The heavier block (**2m**) descends, so its weight wins and the tension opposes it:

$$2mg - T = 2ma$$

The lighter block (**m**) rises, so the tension wins and its weight opposes it:

$$T - mg = ma$$

Step 3 — add to eliminate the tension. Adding the two equations makes **T** cancel:

$$\begin{aligned} (2mg - T) + (T - mg) &= 2ma + ma \\ mg &= 3ma \\ a &= g/3 \end{aligned}$$

The acceleration is one third of **g**, which is plausible: the driving force is the *difference* of the weights, **mg**, and it has to accelerate the *whole* mass, **3m**.

Step 4 — substitute back for the tension. Using the lighter block, because its equation is simpler:

$$T = m(g + a) = m(g + g/3) = 4mg/3$$

Answer: B, $4mg/3$.

Check it against the other block. The heavier block should give the same answer: $2mg - T = 2mg - 4mg/3 = 2mg/3$, and $2ma = 2m(g/3) = 2mg/3$. They agree, so the value is consistent with both equations rather than with just one.

Why the other four are wrong.

- **mg** is the equilibrium answer. It is what you get by setting $a = 0$ above, and it is wrong because the system is visibly accelerating. Note that it is also the *weight of the lighter block*, which is the trap: the tension must be *more* than mg or that block could never rise.
- **$3mg/2$** comes from dividing the driving force by the heavier mass only, $a = mg/2m = g/2$. The driving force has to accelerate both blocks, so the correct divisor is the total mass.
- **$2mg/3$** comes from using $T = m(g - a)$ on the lighter block. That is the formula for a block *descending*; the lighter block is rising, so the tension term adds.
- **$8mg/3$** comes from adding where you should subtract, $T = 2mg + 2ma$. Physically that would mean the string pushes the descending block upward harder than gravity pulls it down, which would make it rise.

The trap. Both mg and $2mg/3$ look like natural answers because each is a weight in the problem. Neither is a tension in a moving system. A quick sanity test settles it: the lighter block accelerates upward, so the net force on it points upward, so $T > mg$. Only $4mg/3$ and $8mg/3$ survive that test, and the second would mean the heavier block accelerates upward too.

Relevant topics: Newton's second law; connected particles; pulley systems; simultaneous equations.

The trap. Answering mg because the tension in a stationary string equals the weight. In a moving system the lighter block is accelerating upward, so T must exceed mg — anything less cannot lift it.

Q2 · A potential divider measured with a voltmeter that is not ideal — **Answer C**

What is being tested. Whether you treat a voltmeter as a measuring device that observes without disturbing, or as what it actually is: a resistor connected across the component.

Step 1 — the meter is part of the circuit. The question gives its resistance, $6.0 \text{ k}\Omega$, which is the signal that it must be included. It sits in parallel with the $3.0 \text{ k}\Omega$ resistor.

Step 2 — combine the parallel pair.

$$R_p = (3.0 \times 6.0) / (3.0 + 6.0) = 18 / 9.0 = 2.0 \text{ k}\Omega$$

Note that the pair is smaller than either member, as any parallel combination must be. That is the physical content of "the meter loads the circuit".

Step 3 — the divider now splits between 6.0 and 2.0.

$$R_{\text{total}} = 6.0 + 2.0 = 8.0 \text{ k}\Omega$$

$$V_{\text{out}} = 12 \times (2.0 / 8.0) = 12 \times 1/4 = 3.0 \text{ V}$$

Step 4 — the reading.

$$V = 3.0 \text{ V}$$

Step 5 — check the direction of the change. With an ideal voltmeter the divider would give $12 \times 3/9 = 4.0 \text{ V}$. The real reading is 3.0 V , which is lower, as it must be: the meter has reduced the resistance of the lower arm, so the lower arm now takes a smaller share of the supply. Any answer above 4.0 V is impossible, which already eliminates one option before any arithmetic.

Answer: C, 3.0 V.

Why the other four are wrong.

- **4.0 V** is the ideal-voltmeter answer, $12 \times 3/(6+3)$. It is the most popular wrong answer because it is the answer to the question you expected.
- **2.0 V** is the parallel resistance $2.0 \text{ k}\Omega$ reported as a voltage. A units check kills it: 2.0 is a resistance, not a potential difference.
- **1.5 V** halves the correct answer, which happens if you halve R_p a second time.
- **6.0 V** is half the supply, which is what a balanced divider would give. Here the two arms are 6.0 and 2.0, nowhere near balanced.

The wider point. Every measurement disturbs what it measures. In Round 0 the disturbance is usually stated as a finite resistance, and the question is really asking whether you will notice. The same idea appears with an ammeter's resistance in series, and with a thermistor whose own resistance changes as it self-heats.

Relevant topics: potential dividers; parallel resistance; real meters; loading effects.

The trap. Computing $12 \times 3/(6+3) = 4.0 \text{ V}$ and never using the $6.0 \text{ k}\Omega$ the question supplied. If a resistance is given, something is meant to be done with it.

Q3 · Dimensional analysis: the terminal speed of a raindrop against its radius — **Answer B**

What is being tested. Whether you can use dimensions as a tool rather than as a memory test. Nothing in the data booklet gives the drag law; it has to be constructed.

Step 1 — build the drag force from its ingredients. Suppose $F = k \rho^a r^b v^c$. Write the dimensions of each:

$$[F] = M L T^{-2}$$

$$[\rho] = M L^{-3} \quad [r] = L \quad [v] = L T^{-1}$$

Matching powers of M, L and T in turn:

$$\begin{aligned} M: \quad 1 &= a \\ T: \quad -2 &= -c \quad \text{so } c = 2 \\ L: \quad 1 &= -3a + b + c = -3 + b + 2 \quad \text{so } b = 2 \end{aligned}$$

So the drag must go as

$$F \propto \rho r^2 v^2$$

This is the familiar result that drag depends on the *cross-sectional area*, which is πr^2 , not on the radius itself. That distinction is the whole question.

Step 2 — at terminal speed, drag balances weight. The drop's weight is

$$W = (4/3) \pi r^3 \rho_w g$$

so it goes as r^3 . Setting drag equal to weight:

$$\rho r^2 v^2 \propto r^3 \rho_w g$$

Step 3 — collect the powers of r . Divide both sides by r^2 :

$$v^2 \propto r \times (\rho_w g / \rho)$$

Step 4 — take the square root. Everything except r is a constant for a given drop and a given air density, so

$$v \propto \sqrt{r}$$

Answer: B, $v \propto \sqrt{r}$.

Why the other four are wrong.

- $v \propto r$ comes from using r in place of the area in the drag law. It is the single most common slip, and the dimensional analysis above is what rules it out: $\rho r v^2$ has dimensions $M L^{-1} T^{-2}$, which is a pressure, not a force.

- $v \propto r^2$ comes from dropping the square root at the end, or from treating the weight as going as r^3 while the drag goes as r .

- $v \propto 1/\sqrt{r}$ inverts the answer. It would mean a large drop falls more slowly than a small one, which is the opposite of what happens: this is why drizzle drifts and large drops sting.

- v independent of r would require the drag and the weight to grow at the same rate in r . They do not — r^2 against r^3 .

The wider point. Whenever a question says a quantity depends only on a list of others, dimensions alone may fix the answer. Try it first: it is often faster than the physics, and it is a check on the physics afterwards. Here it gave both the r^2 in the drag law and the final square root.

Relevant topics: dimensional analysis; drag; terminal velocity; proportional reasoning.

The trap. Treating the drag as proportional to r rather than to the cross-sectional area r^2 . Dimensions settle it immediately: $\rho r v^2$ is a pressure, not a force.

Q4 · An inelastic collision followed by a spring: two conservation laws in a row — **Answer E**

What is being tested. Whether you keep the two conservation laws apart. This question cannot be done with energy alone, and it cannot be done with momentum alone; it needs each one on the right stage.

Step 1 — decide what happens in the collision. The bullet embeds, so bullet and block move off together. Kinetic energy is *not* conserved (that is what "embeds" means), but no external horizontal force acts during the very short collision, so momentum is.

$$m u = (m + 3m) V$$

Step 2 — the common speed.

$$V = m u / (4m) = u/4$$

Three quarters of the speed has gone in one step. That is the price of the inelastic collision, and it is where the kinetic energy is lost.

Step 3 — now energy, for the compression stage. From here on the surface is frictionless and the only force doing work is the spring, so mechanical energy is conserved. All the kinetic energy of the combined mass becomes spring energy at maximum compression:

$$0.5 \times (4m) \times V^2 = 0.5 \times k \times x^2$$

Step 4 — solve for the compression. Cancel the 0.5, substitute $V = u/4$:

$$4m \times (u/4)^2 = k x^2$$

$$4m \times u^2/16 = k x^2$$

$$m u^2 / 4 = k x^2$$

$$x = (u/2) \sqrt{(m/k)}$$

Answer: E, $(u/2)\sqrt{(m/k)}$.

Why the other four are wrong. Each one is the same chain with one step mis-set, which is why they are all dimensionally a length:

- $u\sqrt{(m/k)}$ stores the bullet's *original* kinetic energy, $0.5 m u^2$. That is the energy before the collision, most of which has already been lost as heat and sound.
- $(u/4)\sqrt{(m/k)}$ gets $V = u/4$ right but then stores $0.5 m V^2$, using the bullet's mass alone for the moving mass. The moving mass is $4m$.
- $(u/4)\sqrt{(2m/k)}$ stores $0.5 \times (2m) \times V^2$ — the bullet plus only *one* of the three block masses. You can check the mass directly: with $2m$ the compression comes to $(u/4)\sqrt{(2m/k)}$, which is $1/\sqrt{2}$ of the correct value, and a compression smaller than the true one would leave the block still moving.
- $2u\sqrt{(m/k)}$ assumes the post-collision speed is still u , so no momentum was lost at all. Putting $V = u$ into $x = V\sqrt{(4m/k)}$ gives exactly this.

The trap. The word "embeds" is the instruction. It tells you the collision is perfectly inelastic, so you must run momentum first and energy second. A candidate who writes $0.5 m u^2 = 0.5 k x^2$ has used energy across a stage where energy is not conserved, and will get $u\sqrt{(m/k)}$.

Relevant topics: conservation of momentum; inelastic collisions; elastic potential energy; two-stage problems.

The trap. Applying energy conservation across the collision. The collision is inelastic, so most of the kinetic energy is lost; only momentum survives it. Energy conservation starts *after* the bullet has embedded.

Q5 · Maximum power in a variable load driven by a cell of internal resistance — **Answer D**

What is being tested. Whether you can find a maximum without calculus, and whether you notice that the answer cannot be E^2/r on dimensional grounds alone (that is a power, but it is the power the cell would deliver into a short circuit of zero resistance, where the voltage across the load is zero).

Step 1 — the current. The cell's internal resistance is in series with the load, so the total resistance is $R + r$:

$$I = E / (R + r)$$

Step 2 — the power in the load.

$$P = I^2 R = E^2 R / (R + r)^2$$

Step 3 — tidy it into a shape you can maximise. Divide top and bottom by r^2 and write $x = R/r$:

$$P = (E^2/r) x / (1 + x)^2$$

Everything that depends on R now sits in the factor $x/(1+x)^2$. Maximising P is the same as maximising that factor.

Step 4 — maximise the factor. Expand the denominator:

$$(1 + x)^2 = 1 + 2x + x^2$$

Divide by x :

$$(1 + x)^2 / x = 1/x + 2 + x$$

The factor is largest when this denominator is smallest. Now $1/x + x$ is the sum of two positive numbers whose product is 1 , and the sum of two positive numbers with a fixed product is least when they are equal. So the minimum is at $x = 1$, giving $1 + 2 + 1 = 4$. Therefore

$$x / (1 + x)^2 \leq 1/4$$

The maximum is at $R = r$.

Step 5 — substitute back.

$$P_{\text{max}} = (E^2/r) x (1/4) = E^2 / (4r)$$

Answer: D, $E^2/(4r)$.

Why the other four are wrong.

· E^2/r is the short-circuit power, obtained by setting $R = 0$. But with $R = 0$ there is no load to dissipate anything, so that power is all inside the cell.

· $E^2/(2r)$ comes from assuming half the emf appears across the load — true at $R = r$ — while forgetting that the current is also halved. Using $P = V^2/R$ with $V = E/2$ and $R = r$ gives $E^2/(4r)$; the error is to drop the $/R$.

· $E^2/(8r)$ halves the correct value a second time.

· $E^2 r$ multiplies where it should divide. A quick dimensional check — E^2 is $(\text{volts})^2 = \text{watts} \times \text{ohms}$ — shows that E^2/r is a power and $E^2 r$ is not.

The wider point. Notice the useful sanity check at the start: the power must vanish at both ends of the range of R . At $R = 0$ the voltage across the load is zero, so $P = VI = 0$. As R grows without limit the current tends to zero, so again $P = 0$. A maximum in between is therefore certain, and any option that fails this test is impossible. That kind of reasoning eliminates options before you have done any algebra.

Relevant topics: internal resistance; electrical power; maximisation; inequalities.

The trap. Assuming the maximum power is E^2/r , the short-circuit value. At $R = 0$ there is no potential difference across the load, so the power delivered to it is zero, not a maximum.

Q6 · Estimation: the mass of the air standing on a square metre — **Answer B**

What is being tested. Whether you can turn a pressure into a mass, and whether you know that a pressure is a *force* per unit area. This is an estimation question, so only the power of ten matters.

Step 1 — where the pressure comes from. The air above a patch of ground is held up by that ground. So the force the atmosphere exerts on a square metre is exactly the weight of the column of air above it. This is the step people skip, and it is the whole question.

$$p = W / A$$

Step 2 — the weight of a one square metre column. With $A = 1 \text{ m}^2$:

$$W = p A = 10^5 \text{ N}$$

A hundred thousand newtons per square metre. That is about ten tonnes of force pressing on every square metre, which is why the number is worth pausing over.

Step 3 — turn weight into mass.

$$m = W / g = 10^5 / 10 = 10^4 \text{ kg}$$

Step 4 — sanity check. Ten tonnes per square metre sounds enormous until you picture it: the column reaches the top of the atmosphere, roughly 10 km of air at about 1 kg m^{-3} near the ground. A 10 km column at that density is $10^4 \text{ m} \times 1 \text{ kg m}^{-3} = 10^4 \text{ kg m}^{-2}$. The two routes agree, which is a good sign.

Answer: B, 10^4 kg .

Why the other four are wrong.

· 10^5 kg is the pressure in pascals quoted as a mass. It is the mistake of stopping one step early.

- 10^6 kg comes from multiplying by g instead of dividing. A weight is *bigger* than the mass that produces it, so multiplying makes the answer move the wrong way.
- 10^3 kg divides by 100, which is what you get by using $g = 100$.
- 10^7 kg is two powers of ten out.

The wider point. Estimation questions are marked on the power of ten, and the way to be safe is to keep track of the physical meaning of each quantity rather than the symbols. "Pressure is a weight per unit area" is the fact being tested; once you have it, the arithmetic is a single division.

Relevant topics: pressure; weight; estimation; orders of magnitude.

The trap. Quoting 10^5 kg because the pressure is 10^5 Pa . A pascal is a newton per square metre, so the pressure gives you a weight, and you must still divide by g to get a mass.

Q7 · A projectile passing the same height on the way up and on the way down — **Answer E**

What is being tested. Whether you see that "twice" means "two roots of one equation", and whether you can take a difference of roots without grinding out both times separately.

Step 1 — write the displacement. Taking upward as positive, and remembering that gravity acts downward:

$$s = ut - 0.5gt^2$$

Step 2 — impose the condition. The ball is at height h when $s = h$:

$$h = ut - 0.5gt^2$$

Rearranged into standard form:

$$0.5gt^2 - ut + h = 0$$

This is a quadratic in t , so it has two roots. That is the whole content of "it passes the point twice": one root is the time on the way up, the other the time on the way down.

Step 3 — solve the quadratic. With $a = 0.5g$, $b = -u$, $c = h$:

$$t = [u \pm \sqrt{u^2 - 2gh}] / g$$

The two roots are $t_1 = (u - \sqrt{u^2 - 2gh})/g$ and $t_2 = (u + \sqrt{u^2 - 2gh})/g$.

Step 4 — subtract. The interval is

$$t_2 - t_1 = 2\sqrt{u^2 - 2gh} / g$$

Notice that u has cancelled entirely from the difference. That is worth pausing on: the interval depends on h and g , but not on how hard the ball was thrown — as long as it is thrown hard enough to get there at all.

Answer: E, $2\sqrt{u^2 - 2gh}/g$.

Why the other four are wrong.

- $\sqrt{(u^2 - 2gh)}/g$ is exactly half the answer. It is the time from the apex down to h , because at the apex the speed is zero and the time to fall a distance $(u^2/2g - h)$ is $\sqrt{(2(u^2/2g - h)/g)} = \sqrt{(u^2 - 2gh)}/g$. By symmetry that is also the time from h up to the apex, so the full interval is twice it.
- u/g is the time to reach the highest point — the middle of the interval, not its length.
- $2\sqrt{(2gh)}/g$ drops the u^2 term, which is what you get by treating the ball as simply dropped from height h . That would require the ball to have zero speed at h , in which case it would only be there once.
- $2u/g$ is the whole flight time, ground to ground. It is the special case $h = 0$, and it is the largest the interval can ever be.

A quick numerical check. Take $u = 5 \text{ m/s}$, $g = 10 \text{ N kg}^{-1}$ and $h = 0.8 \text{ m}$. Then $u^2 - 2gh = 25 - 16 = 9$, so the interval is $2 \times 3 / 10 = 0.6 \text{ s}$. Solving the quadratic directly gives roots 0.8 s and 0.2 s , whose difference is indeed 0.6 s . The formula and the two-times route agree.

Relevant topics: constant acceleration; quadratic equations; symmetry of projectile motion.

The trap. Answering $\sqrt{(u^2 - 2gh)}/g$ — half the interval. That is the time from the apex down to h , and by symmetry it is also the time from h up to the apex, so the gap between the two passages is twice it.

Q8 · A horizontal beam held by an angled stay: moments about the hinge — **Answer C**

What is being tested. Two things at once: that the weight of a uniform beam acts at its midpoint, and that an angled force only produces a moment through the component perpendicular to the beam.

Step 1 — choose the pivot. Take moments about the hinge. The beam is in equilibrium, so the total moment about *any* point is zero, and we are free to pick the most convenient one. The hinge is convenient because the force the hinge exerts acts there: its lever arm is zero, so it contributes nothing and never has to be found.

Step 2 — the moment of the weight. The beam is uniform, so its centre of gravity is at its midpoint, a distance $L/2 = 2.0 \text{ m}$ from the hinge. The weight is

$$W = m g = 20 \times 10 = 200 \text{ N}$$

acting downward, so the moment is

$$\text{moment of } W = 200 \times 2.0 = 400 \text{ N m} \quad (\text{clockwise})$$

Step 3 — the moment of the tension. The stay makes 30° with the beam. Only the component of T perpendicular to the beam produces a moment, and that component is $T \sin 30^\circ$. It acts at the far end, a distance $L = 4.0 \text{ m}$ from the hinge:

$$\text{moment of } T = T \sin 30^\circ \times 4.0 \quad (\text{anticlockwise})$$

Step 4 — equate and solve.

$$\begin{aligned} T \sin 30^\circ \times 4.0 &= 400 \\ T \times 0.5 \times 4.0 &= 400 \\ 2.0 T &= 400 \\ T &= 200 \text{ N} \end{aligned}$$

Answer: C, 200 N.

Why the other four are wrong.

- **100 N** is $mg/2$: the stay treated as vertical, so the tension is just half the weight. The angle has been ignored.
- **50 N** multiplies by $\sin 30^\circ$ instead of dividing by it, which moves the answer the wrong way. A quick physical check settles it: the stay pulls at 30° , so it must pull *harder* than a vertical stay would, not less.
- **115 N** resolves with $\cos 30^\circ$ instead of $\sin 30^\circ$. The perpendicular component of a force at angle θ to the beam is $T \sin \theta$; the cosine gives the component *along* the beam, which produces no moment about the hinge at all.
- **400 N** takes moments about the far end, so the whole weight acts at the full length L rather than at $L/2$.

The wider point. In every statics question, the first decision is where to put the pivot, and the right answer is "somewhere that removes an unknown". Here that is the hinge. If the question had also asked for the hinge force, you would then resolve horizontally and vertically to find it — the moment equation would already have done the hard part.

Relevant topics: moments; equilibrium; centre of gravity; resolving forces.

The trap. Using the whole weight at the far end. A uniform beam's weight acts at its midpoint, so its moment about the hinge is $mg \times L/2$, not $mg \times L$.

Q9 · The speed at which a car leaves the road on a hump-backed bridge — **Answer A**

What is being tested. Whether you can identify the "just leaving" condition. It is not a new force appearing; it is an existing force disappearing.

Step 1 — the forces at the top of the arc. Two forces act on the car at the top: its weight mg downward, and the normal reaction N from the road upward. The centre of the circular arc is *below* the car, so the inward (centripetal) direction is downward, and the net inward force is

$$F = mg - N$$

Step 2 — that net force is the centripetal force.

$$mg - N = m v^2 / r$$

Read that equation physically. As v grows, the right-hand side grows, so N must shrink. The road has to push less and less hard the faster the car goes.

Step 3 — the limiting condition. The road can only push, never pull. So the smallest N can be is zero, and "just leaving the road" is exactly $N = 0$:

$$mg = m v^2 / r$$

Step 4 — solve and evaluate. The mass cancels, which is worth noticing: the answer does not depend on how heavy the car is.

$$v^2 = g r = 10 \times 20 = 200$$

$$v = \sqrt{200} \approx 14 \text{ m/s}$$

Answer: A, 14 m/s.

Why the other four are wrong.

- **10 m/s** comes from setting $N = mg/2$ as the losing-contact condition, i.e. from assuming the car leaves when the reaction has merely halved. It leaves when the reaction reaches zero.
- **20 m/s** uses the diameter $2r = 40 \text{ m}$ in place of the radius. It is worth checking which one the geometry gives you: the radius is the distance from the car to the centre of the arc, which is 20 m.
- **200 m/s** is gr , the square root forgotten. A units check catches it: gr has units $\text{m}^2 \text{s}^{-2}$, not m s^{-1} .
- **400 m/s** is $2gr$, so it uses the diameter *and* forgets the root.

The wider point. The same structure appears in a ball whirled in a vertical circle (the string goes slack at the top), in a satellite at the surface, and in a bucket of water swung overhead. In every case, the quantity that can only push or only pull reaches its limit at zero, and setting it to zero gives the threshold.

Relevant topics: circular motion; centripetal force; normal reaction; limiting conditions.

The trap. Looking for a new force to appear. Nothing new happens when the car leaves the road — the normal reaction, which can only push, simply reaches zero. Set $N = 0$, not $N = mg/2$.

Q10 · Three successive resonances of a pipe closed at one end — **Answer D**

What is being tested. Whether you know which harmonics a closed pipe can produce. A pipe open at both ends gives all of them; a pipe closed at one end gives only the odd ones, and that single fact answers this question.

Step 1 — what a closed pipe allows. The closed end must be a node (the air cannot move there) and the open end an antinode (the air is free to move). The shortest pattern that fits is a quarter wavelength, and every longer one adds a half wavelength:

$$L = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$$

$$f = f_0, 3f_0, 5f_0, \dots$$

where f_0 is the fundamental. So the allowed frequencies are the *odd* multiples of the fundamental.

Step 2 — the gap between neighbours. Consecutive allowed frequencies are f_0 , $3f_0$, $5f_0$, so the gap between any two neighbours is

$$3f_0 - f_0 = 2f_0$$

Step 3 — read the gap off the data. The three given frequencies are 300, 500 and 700 Hz, and the question says there is nothing in between. The gap is

$$500 - 300 = 200 \text{ Hz} \quad (\text{and } 700 - 500 = 200 \text{ Hz})$$

So $2f_0 = 200 \text{ Hz}$ and

$$f_0 = 100 \text{ Hz}$$

Step 4 — check. If the fundamental is 100 Hz the odd harmonics are 100, 300, 500, 700, 900 Hz. The three quoted values are exactly the 3rd, 5th and 7th. The 100 Hz resonance itself is simply not among the three that were measured, which is perfectly consistent with the question as worded.

Answer: D, 100 Hz.

Why the other four are wrong.

- **300 Hz** is the lowest frequency given. It is a resonance, but it is the third harmonic, not the fundamental. This is the most tempting wrong answer, because it is the smallest number on the page.
- **200 Hz** is the gap. It is $2f_0$, not f_0 .
- **150 Hz** averages 100 and 200; it is not a frequency this pipe can produce at all.
- **50 Hz** divides the gap by 4, which would be right for a pipe open at *both* ends, where the gap is f_0 ... — no, for an open pipe the gap between successive resonances is f_0 itself, so that error does not even arise there. It is simply a factor of two out from the correct answer.

The wider point. Whenever a question gives you a *sequence* of resonances and asks for the fundamental, use the spacing. It is more robust than trying to identify which harmonic each frequency is, and it works even when the lowest resonances have not been measured.

Relevant topics: stationary waves in pipes; harmonics; boundary conditions; resonance.

The trap. Answering 300 Hz because it is the smallest frequency given. A closed pipe produces only odd harmonics, and 300 Hz is the third harmonic of a 100 Hz fundamental.

Q11 · The minimum refractive index for total internal reflection inside a prism — **Answer C**

What is being tested. Whether you can combine a geometric fact with an optical one. The geometry is supplied by the shape of the prism; the optics by the critical-angle condition.

Step 1 — the first face does nothing. The ray enters normally, meaning perpendicular to the surface. A ray along the normal is undeviated by refraction, so it carries on in a straight line inside the glass. No Snell's law calculation is needed here, and trying to do one is a waste of the two minutes.

Step 2 — where the ray arrives. The prism is a right-angled isosceles triangle, so its two acute angles are 45° . A ray travelling along one of the short directions meets the long face at 45° to the normal. This is the angle of incidence at the glass-air boundary:

$$\theta = 45^\circ$$

Step 3 — the condition for total internal reflection. Light inside a medium of refractive index n striking a boundary with air is totally internally reflected if the angle of incidence exceeds the critical angle θ_c , where

$$\sin \theta_c = 1/n$$

So the requirement is $\sin 45^\circ > 1/n$, or equivalently

$$n > 1 / \sin 45^\circ$$

Step 4 — the smallest value. The question asks for the smallest n , which is the equality case:

$$n_{\min} = 1 / \sin 45^\circ = 1 / (1/\sqrt{2}) = \sqrt{2} \approx 1.4$$

Answer: C, $\sqrt{2}$.

Why the other four are wrong.

- **1.5** is a typical glass, and it does give total internal reflection here (since $1.5 > \sqrt{2}$). It is wrong only because the question asks for the *smallest* index that works. Reading the question as "will it reflect?" instead of "what is the threshold?" is the trap.
- **4/3** is the refractive index of water, quoted because it is a familiar number. It is *below* $\sqrt{2}$, so a prism made of it would leak the ray out at the long face.
- **1.2** uses $\sin 55^\circ$ in place of $\sin 45^\circ$, and is well below the threshold.
- **$\sqrt{3}$** uses 30° instead of 45° . A 30-60-90 prism would indeed have a threshold of $1/\sin 30^\circ = 2$, not $\sqrt{3}$, so this option is wrong twice over.

The wider point. Prisms that turn a beam through 90° or 180° by total internal reflection are common in optical instruments, and they rely on exactly this: the glass must be chosen so that 45° exceeds the critical angle. Ordinary crown glass at $n = 1.5$ clears the threshold with a little room to spare, which is why such prisms are made of it.

Relevant topics: total internal reflection; critical angle; prism geometry; threshold conditions.

The trap. Choosing **1.5** because that is a typical glass and it does reflect. The question asks for the smallest index that works, which is the equality case $1/\sin 45^\circ = \sqrt{2}$.

Q12 · The resistance of a filament lamp at one point on its characteristic — **Answer D**

What is being tested. A single distinction that decides a whole family of questions: on a graph of V against I , the resistance is the *ratio* V/I , which is the slope of the line drawn from the origin to the point — not the gradient of the curve at the point.

Step 1 — what resistance means at a point. Resistance is defined by $R = V/I$. That is a statement about a pair of values, so it applies at one point on the graph, not along a tangent.

Step 2 — the two different slopes. The gradient of the curve is dV/dI , which has units of ohms too but is a different quantity. For a straight line through the origin they coincide, which is exactly why the distinction is easy to miss. For a curved characteristic they do not.

Step 3 — read the point and divide. The marked point is at $V = 6.0 \text{ V}$ and $I = 0.40 \text{ A}$, so

$$R = V / I = 6.0 / 0.40 = 15 \, \Omega$$

Step 4 — check it is sensible. A 6.0 V lamp drawing 0.40 A has a resistance of $15 \, \Omega$, which is a realistic value for a small filament lamp when it is hot. Note also that this is much larger than the lamp's resistance when cold, which is the reason its characteristic curves over rather than being straight.

Answer: D, 15 Ω .

Why the other four are wrong.

- **2.4 Ω** is $V \times I$, the product instead of the ratio. Units catch it: volts times amperes is watts, not ohms.
- **6.0 Ω** divides by **1.0 A**, i.e. the number on the axis rather than the current at the point.
- **7.5 Ω** divides by **0.80 A**, reading the wrong gridline.
- **24 Ω** divides by **0.25 A**, again a misread gridline.

The wider point. The same distinction appears in a stress-strain graph, where Young's modulus is the gradient of the linear part rather than a ratio of coordinates, and in a velocity-time graph, where the gradient is acceleration but the ratio of coordinates is average velocity. Ask which of the two the question wants before you start reading numbers.

Relevant topics: resistance; current-voltage characteristics; non-ohmic conductors; reading graphs.

The trap. Taking the gradient of the curve. On a curved $V - I$ characteristic the gradient dV/dI is not the resistance; the resistance at a point is V/I , the slope of the line from the origin.

Q13 · Thermal equilibrium between a hot metal block and cooler water — **Answer E**

What is being tested. Whether you identify the correct weighting. The answer is a weighted average of 200 °C and 20 °C, and the weights are the heat capacities mc , not the masses and not the specific heat capacities.

Step 1 — the principle. In an insulated container no heat escapes, so whatever the block loses the water gains:

$$\text{heat lost by block} = \text{heat gained by water}$$

Step 2 — write each side. Let the final temperature be θ . The block cools from 200 °C and the water warms from 20 °C:

$$m_{\text{block}} c_{\text{block}} (200 - \theta) = m_{\text{water}} c_{\text{water}} (\theta - 20)$$

It is worth computing each heat capacity separately, because that is the quantity that actually does the weighting:

$$\begin{aligned} C_{\text{block}} &= 0.20 \times 500 = 100 \text{ J K}^{-1} \\ C_{\text{water}} &= 0.10 \times 4000 = 400 \text{ J K}^{-1} \end{aligned}$$

Step 3 — substitute and collect.

$$\begin{aligned} 100 (200 - \theta) &= 400 (\theta - 20) \\ 20000 - 100\theta &= 400\theta - 8000 \\ 28000 &= 500\theta \end{aligned}$$

Step 4 — solve.

$$\theta = 28000 / 500 = 56 \text{ }^{\circ}\text{C}$$

Answer: E, 56 °C.

Check the answer is between the two starting values. It must be: energy flows from hot to cold, so the final temperature cannot be above 200 °C or below 20 °C. That single test eliminates two of the options immediately. It is also much closer to 20 °C than to 200 °C, which is right, because the water's heat capacity is four times the block's.

Why the other four are wrong.

- **110 °C** is the plain average $(200+20)/2$. It would be correct only if the two heat capacities were equal, and they are not — 100 against 400.
- **140 °C** weights by mass alone, giving $(0.20 \times 200 + 0.10 \times 20)/0.30$. The specific heat capacities matter as much as the masses.
- **40 °C** uses the specific heat capacities without the masses. Both factors are needed; only their product has the right units, J K⁻¹.
- **20 °C** leaves the water unchanged, as though dropping in a hot block did nothing.

The wider point. "Weighted average" questions are common and the weight is easy to get wrong. The safe method is always the same: write the two heat quantities in full, $mc\Delta T$ on each side, and let the algebra do the weighting for you.

Relevant topics: specific heat capacity; thermal equilibrium; conservation of energy.

The trap. Answering 110 °C by averaging the two temperatures. The final temperature is weighted by the heat capacities mc , and here the water's is four times the block's, so the answer sits far nearer 20 °C.

Q14 · Alpha decay: how the released energy is divided between the two products — **Answer A**

What is being tested. Whether you realise that the energy released in a decay is *shared*, and whether you can work out the sharing from momentum conservation alone.

Step 1 — momentum. The nucleus is at rest, so its momentum is zero. Momentum is conserved, so the two products must have equal and opposite momenta:

$$m_{\alpha} v_{\alpha} = M_{\text{recoil}} v_{\text{recoil}}$$

Step 2 — from momentum to energy. Kinetic energy can be written in terms of momentum:

$$KE = p^2 / (2m)$$

Since p is the same for both products, the kinetic energies are *inversely* proportional to the masses:

$$KE_{\alpha} / KE_{\text{recoil}} = M_{\text{recoil}} / m_{\alpha}$$

The lighter particle moves faster and carries more of the energy. That is the physical content of the question.

Step 3 — put in the masses. An alpha particle is a helium nucleus, mass number 4. The recoil nucleus keeps the remaining nucleons:

$$m_{\alpha} \propto 4 \qquad M_{\text{recoil}} \propto A - 4$$

So the energies are in the ratio $(A - 4) : 4$. The alpha's share of the total is therefore

$$KE_{\alpha} = E \times (A - 4) / ((A - 4) + 4) = E (A - 4)/A$$

Step 4 — check the extremes. For a very heavy nucleus, A is large and $(A-4)/A$ is close to 1, so the alpha takes nearly all the energy and the heavy recoil takes almost none — which is right, since a heavy nucleus barely moves. This is why alpha energies measured in a detector are sharply defined even though the decay energy is shared: the sharing is nearly total and depends only on A .

Answer: $A, E(A - 4)/A$.

A numerical check. Take $A = 6$, so the recoil has mass number 2, and let $E = 6$ units. The formula gives $6 \times 2/6 = 2$. Directly: the energies are in the ratio $2 : 4$, i.e. the alpha gets two thirds of nothing of the sort — rather, the alpha's share is $(A-4)/A = 2/6$, so 2 units. The two routes agree.

Why the other four are wrong.

- $4E/A$ gives the alpha the share proportional to its own mass. That is the classic inversion, and it says the alpha carries the *smaller* part — the opposite of what equal momenta require.
- $4E/(A - 4)$ inverts the ratio outright and is larger than E for any $A < 8$, which is impossible.
- $E/2$ would need the two fragments to have equal masses. An alpha particle has mass number 4, so that requires $A = 8$ exactly.
- E ignores the recoil altogether, which would violate momentum conservation.

Relevant topics: alpha decay; conservation of momentum; kinetic energy in terms of momentum; mass number.

The trap. Giving the alpha particle the share proportional to its own mass. Equal and opposite momenta mean the LIGHTER fragment carries the larger kinetic energy, so the alpha takes $(A-4)/A$, not $4/A$.

Q15 · Small change: the resistance of a wire stretched at constant volume — **Answer B**

What is being tested. Whether you notice that stretching a wire changes *two* things at once. A candidate who only thinks about the length answers x ; the cross-section has other ideas.

Step 1 — the resistance formula.

$$R = \rho L / A$$

The resistivity ρ is a property of the metal and does not change when the wire is stretched. So R depends on L and A , and both of those change.

Step 2 — what constant volume does. The volume of the wire is its length times its cross-sectional area, and it is fixed:

$$L A = V = \text{constant}$$

So if the length grows, the area must shrink in proportion:

$$A = V / L, \quad \text{so} \quad A \propto 1/L$$

Step 3 — eliminate the area. Substituting into the resistance formula:

$$R = \rho L / (V/L) = (\rho/V) L^2$$

So the resistance is proportional to the *square* of the length. This is the key step, and it is worth remembering as a fact in its own right: a wire stretched at constant volume has resistance proportional to L^2 .

Step 4 — apply the small change. The new length is $L(1 + x)$, so the new resistance is

$$R' = R (1 + x)^2 = R (1 + 2x + x^2)$$

Since x is small, x^2 is negligible beside $2x$, so

$$R' \approx R (1 + 2x), \text{ giving } \Delta R / R \approx 2x$$

Answer: B, $2x$.

Why the other four are wrong.

- x uses the length alone, ignoring that the area has shrunk. Both effects increase the resistance, so the answer must be bigger than x .
- $x/2$ treats the area change as reducing the effect. It does not: a thinner wire has more resistance.
- $4x$ squares the factor of two, which happens if you write $(1+x)^2 \approx (1+2x)^2$ or otherwise double-count.
- 0 would require the two effects to cancel, i.e. the area to fall as fast as the length rises in the resistance formula. But the area enters as $1/A$, so a fractional fall in area gives the same fractional rise in resistance as the same fractional rise in length — they add, they do not cancel.

The wider point. "Stretched at constant volume" and "stretched so the length doubles" are different statements, and the difference is the area. Whenever a wire is stretched, ask whether the volume is being held fixed; if it is, the resistance goes as the square of the length.

Relevant topics: resistivity; small changes; binomial approximation; proportional reasoning.

The trap. Answering x by treating the resistance as proportional to the length. The volume is constant, so the cross-section shrinks as the length grows, and $R \propto L^2$ — the fractional change is $2x$.

Q16 · Choosing the velocity-time graph of a bouncing ball — **Answer E**

What is being tested. Whether you can read the physics off the *shape* of a graph. Two features decide it, and they are independent of each other.

Step 1 — what the slope means. On a velocity-time graph the gradient is the acceleration. While the ball is in the air the only force on it is its weight, so its acceleration is g downward — the same value on the way up as on the way down, and the same at the top of the flight as anywhere else. In particular the acceleration is **not** zero at the highest point; only the velocity is.

So every section of the graph between bounces must be a **straight line**, and every one of those straight lines must have the **same** slope. A curve anywhere means a changing acceleration, which would need

the force to change.

$$v = u + a t, \text{ with } a = -g \text{ throughout the flight}$$

Step 2 — what happens at the bounce. The ball hits the floor moving downward and leaves it moving upward, so the velocity reverses sign. The magnitude depends on how much energy the bounce returns: since the ball rises to a smaller height, the rebound speed is smaller.

Using $v^2 = 2gh$, a bounce to a quarter of the height returns half the speed:

$$v_{\text{rebound}} / v_{\text{arrival}} = \sqrt{(h_{\text{rebound}} / h_{\text{fall}})}$$

So each bounce gives a smaller jump, and the peak speed reached in each flight is smaller than the one before.

Step 3 — assemble the shape. Starting from rest: a straight line of slope $-g$ going negative; a jump up to a smaller positive value; a straight line of slope $-g$ back down to zero at the top of the bounce; then the pattern repeats with smaller peaks.

Answer: E. It is the only option with straight sections of constant slope *and* shrinking peaks.

Why the other four are wrong.

- **A** has the ball leave the floor at exactly the speed it arrived with. That is a perfectly elastic bounce, so it would rise to the same height — contradicting the question.
- **B** has the slope steepening, which would mean the acceleration grows as the ball falls. Gravity does not do that.
- **C** has the ball start off moving upward. It is released from rest, so its velocity starts at zero.
- **D** gets the shape right but gives every bounce the same peak speed, so the ball would rise to the same height each time.

The wider point. A velocity-time graph answers two questions at once, and it is worth asking them separately. The *slope* tells you about force and acceleration; the *value* tells you about speed and direction. Most graph-choice questions are decided by one or the other, and the best ones by both.

Relevant topics: velocity-time graphs; free fall; bounces and energy loss; interpreting gradients.

The trap. Choosing the graph with equal peaks. The ball does not return to its starting height, so the speed after each bounce is smaller. Equally, any graph with a *curved* section is wrong: gravity gives a constant acceleration, so every straight section has the same slope.

Q17 · Limiting friction: the hanging mass that just starts to move a block — **Answer D**

What is being tested. Whether you include both resisting forces. The block is held back by its own weight sliding down the slope *and* by friction, and the hanging mass must overcome both.

Step 1 — resolve the block's weight. The plane has sine 0.60 and cosine 0.80, which is the 3-4-5 triangle, so the arithmetic stays clean. The weight of the block is

$$W = 5.0 \times 10 = 50 \text{ N}$$

Resolving along and perpendicular to the plane:

$$W_{\text{parallel}} = 50 \times 0.60 = 30 \text{ N} \quad (\text{down the slope})$$

$$W_{\text{perpendicular}} = 50 \times 0.80 = 40 \text{ N} \quad (\text{into the plane})$$

Step 2 — the normal reaction and the limiting friction. Perpendicular to the plane nothing is accelerating, so the normal reaction balances the perpendicular component:

$$N = 40 \text{ N}$$

The limiting friction follows from the coefficient:

$$F = \mu N = 0.50 \times 40 = 20 \text{ N}$$

Its direction opposes the motion that is about to start, so it acts **down** the slope, the same way as the weight component.

Step 3 — the condition for just moving. The string's tension equals the hanging weight, mg . "Just starts to move" means the tension exactly balances the two forces resisting motion:

$$mg = W_{\text{parallel}} + F$$

$$10m = 30 + 20 = 50$$

Step 4 — solve.

$$m = 5.0 \text{ kg}$$

Answer: D, 5.0 kg.

Why the other four are wrong.

- **3.0 kg** balances only the weight component, giving $mg = 30$. Friction has been left out.
- **2.0 kg** balances only friction, giving $mg = 20$. The slope component has been left out.
- **4.0 kg** uses the normal reaction, 40 N, as the force to be balanced. The normal reaction acts into the plane, not along it, so it cannot be balanced by a pull along the plane at all — it is a coincidence of these numbers that it lands on a plausible value.
- **7.0 kg** adds friction twice, $mg = 30 + 20 + 20$.

The wider point. Two checks make this kind of question safe. First, draw the free-body diagram and count the forces along the plane: there should be three (tension, weight component, friction). Second, ask which way friction points — it opposes impending motion, so if the block is about to move *up* the slope, friction acts *down* it. Getting that direction wrong turns a sum into a difference and produces a plausible but wrong answer.

Relevant topics: forces on an inclined plane; limiting friction; resolving vectors; equilibrium conditions.

The trap. Balancing only the component of the block's weight along the slope. Friction acts down the slope too, so the hanging mass must overcome $mg \sin\theta + \mu mg \cos\theta$, not just the first term.

What is being tested. Whether you can go from an acceleration requirement to a period, through the angular frequency. Three quantities are involved and it is easy to lose track of which is which.

Step 1 — what "apparent gravity" means. An astronaut standing on the inside of the rim is travelling in a circle. To stay in that circle they need a centripetal acceleration of $\omega^2 r$ directed towards the axis. What they feel as weight is the floor pushing them inward, which is exactly the force providing that acceleration. So the apparent gravity is the centripetal acceleration:

$$a = \omega^2 r$$

Step 2 — impose the requirement. We want $a = g$ at the rim, where $r = 50 \text{ m}$:

$$\begin{aligned} g &= \omega^2 r \\ 10 &= \omega^2 \times 50 \\ \omega^2 &= 0.20 \end{aligned}$$

Step 3 — angular frequency to period.

$$\begin{aligned} \omega &= \sqrt{0.20} \approx 0.45 \text{ rad/s} \\ T &= 2\pi / \omega \approx 6.28 / 0.45 \approx 14 \text{ s} \end{aligned}$$

An equivalent route is $T = 2\pi\sqrt{r/g}$, which some people find easier to remember because it goes straight from the geometry to the period.

Answer: C, 14 s.

Check the answer physically. A period of 14 s is about seven rotations a minute. That is slow enough to be comfortable and fast enough to be useful — which is roughly why proposals of this kind quote periods of this order. If your answer had been a fraction of a second the crew would be centrifuged; if it had been hours the "gravity" would be imperceptible.

Why the other four are wrong.

- **1.4 s** comes from $\omega = g/r = 0.20$ used directly as an angular frequency without taking the square root, which makes ω far too small and T far too short.
- **2.8 s** is $2\pi\sqrt{g/r}$, the ratio the wrong way up.
- **20 s** uses the diameter 100 m instead of the radius. The centripetal acceleration uses the distance from the axis, which is the radius.
- **31 s** is $2\pi r/g$, with the square root dropped. A units check catches it: r/g has units of s^2 , not s.

The wider point. Notice that the mass of the astronaut never appeared, and neither did the mass of the station. Apparent gravity set by rotation is independent of mass, which is why a station of any size can be spun to give $1g$ at a chosen radius.

Relevant topics: circular motion; centripetal acceleration; angular frequency and period; apparent weight.

The trap. Solving $\omega = g/r$ and stopping. That gives ω^2 , not ω ; the square root is easy to lose, and it changes the answer by a factor of about three.

Q19 · Strain energy per unit volume as the area under a stress-strain graph — **Answer B**

What is being tested. Whether you know what the two features of a stress-strain graph mean. The gradient gives one quantity and the area gives another, and they are easy to confuse because both are read off the same curve.

Step 1 — which feature is wanted. Work done on a sample per unit volume is force times distance per unit volume, which is stress times strain — an area under the curve, not a slope. So the answer is the area.

$$\text{energy per unit volume} = \int \sigma \, d\varepsilon$$

Step 2 — the shape of the region. Up to the end of the straight-line region the graph is a straight line through the origin, so the area beneath it is a triangle:

$$\text{area} = 0.5 \times \text{base} \times \text{height}$$

Step 3 — identify base and height. The horizontal axis is strain, so the base is the strain; the vertical axis is stress, so the height is the stress:

$$\begin{aligned} \text{base} &= 1.5 \times 10^{-3} \\ \text{height} &= 3.0 \times 10^8 \text{ Pa} \end{aligned}$$

Step 4 — evaluate.

$$\begin{aligned} \text{area} &= 0.5 \times 3.0 \times 10^8 \times 1.5 \times 10^{-3} \\ &= 0.5 \times 4.5 \times 10^5 \\ &= 2.25 \times 10^5 \text{ J m}^{-3} \end{aligned}$$

To two significant figures this is $2.3 \times 10^5 \text{ J m}^{-3}$.

Answer: B, $2.3 \times 10^5 \text{ J m}^{-3}$.

Why the other four are wrong.

- 4.5×10^5 is the full product $\sigma\varepsilon$, the factor of a half forgotten. That product is the area of the whole rectangle, not of the triangle under the line.
- 1.1×10^5 divides by four instead of by two, halving twice.
- 2.0×10^{11} is the gradient, $\sigma/\varepsilon = 3.0 \times 10^8 / 1.5 \times 10^{-3} = 2.0 \times 10^{11} \text{ Pa}$. That is Young's modulus, a perfectly sensible quantity that the question did not ask for. Its units give it away: pascals, not joules per cubic metre.
- 4.5×10^2 has the right structure but the powers of ten are wrong by a factor of a million.

The wider point. When a question gives you a graph, the first question to ask is whether it wants the gradient or the area, because the two answer different physics. Gradient of a displacement-time graph is velocity; area of a velocity-time graph is displacement. Gradient of a stress-strain graph is Young's modulus; area is energy density. Getting into that habit removes a whole class of errors.

Relevant topics: stress and strain; Young's modulus; strain energy; interpreting graphs.

The trap. Reporting the gradient. σ/ϵ gives Young's modulus in pascals; the strain energy per unit volume is the *area* under the curve, in joules per cubic metre. The units tell you which one you have computed.

Q20 · The highest diffracted order a grating can produce — **Answer A**

What is being tested. Whether you know what stops the orders, and whether you round the right way once you have found the limit.

Step 1 — the grating spacing. The ruling density is given per millimetre, so convert to a spacing in metres:

$$d = 1 \text{ mm} / 500 = 1.0 \times 10^{-3} \text{ m} / 500 = 2.0 \times 10^{-6} \text{ m}$$

Two micrometres between adjacent slits. It is worth converting the wavelength too, since mixing nanometres with millimetres is a standard way to be out by a factor of a million:

$$\lambda = 600 \text{ nm} = 6.0 \times 10^{-7} \text{ m}$$

Step 2 — the grating equation. A bright beam appears at angle θ when the path difference between neighbouring slits is a whole number of wavelengths:

$$d \sin \theta = n \lambda$$

Step 3 — the cap on n. Rearranged,

$$\sin \theta = n \lambda / d$$

The left-hand side is a sine, and a sine cannot exceed one. So

$$n \lambda / d \leq 1, \text{ giving } n \leq d / \lambda$$

Step 4 — evaluate and round the right way.

$$d / \lambda = (2.0 \times 10^{-6}) / (6.0 \times 10^{-7}) = 3.3$$

So n can be 1, 2 or 3 but not 4 — the fourth order would need $\sin \theta = 4\lambda/d = 4 \times 0.30 = 1.20$, and a sine cannot reach that. The answer is therefore 3.

Answer: A, 3.

Why the other four are wrong.

- **4** rounds 3.3 to the nearest whole number. The inequality is $n \leq 3.3$, so 3.3 is a ceiling, not a target: the fourth order does not exist.
- **6** counts the three orders on each side of the straight-through direction. Those beams are real — but the question asks for the highest *order*, which is a label for a beam direction, not a count of beams.
- **7** adds the zero order to that count. The zero order is the undiffracted beam and is not usually counted as a diffracted order at all.
- **12** doubles the 6.

The wider point. This is a "count the possibilities" question, and the counting is where most of the marks are lost. Two things to be careful about: convert every length to the same unit before dividing, and remember that a limit expressed as an inequality rounds down, not to the nearest.

Relevant topics: diffraction gratings; the grating equation; orders of diffraction; limiting conditions.

The trap. Rounding $d/\lambda = 3.3$ up to 4. The condition is an inequality — no sine exceeds 1 — so the answer is the largest whole number that satisfies it, which is 3.

Q21 · The acceptance angle of a step-index optical fibre — **Answer B**

What is being tested. Whether you can handle two boundaries in one ray path, and whether you see the geometry that links them.

Step 1 — what traps the light. Inside the fibre the ray bounces along the boundary between core and cladding. It stays in the core if it strikes that boundary at more than the critical angle, where

$$\sin \theta_c = n_2 / n_1 = 1.2 / 1.5 = 0.8$$

So the ray must meet the wall at an angle of at least θ_c measured from the normal to the wall — and the normal to the wall is a radius, perpendicular to the axis.

Step 2 — the geometry. Let the ray make an angle α with the fibre's axis. The angle it makes with the wall's normal is then $90^\circ - \alpha$. The trapping condition is

$$90^\circ - \alpha \geq \theta_c$$

so the steepest ray that survives has $90^\circ - \alpha = \theta_c$, i.e. $\alpha = 90^\circ - \theta_c$. Taking the sine:

$$\sin \alpha = \sin(90^\circ - \theta_c) = \cos \theta_c$$

Step 3 — Snell's law at the end face. Light comes from air (index 1) into the core (index n_1), and the angle inside is α :

$$\sin \theta_a = n_1 \sin \alpha = n_1 \cos \theta_c$$

Step 4 — eliminate the internal angle. Since $\sin \theta_c = n_2/n_1$,

$$\cos \theta_c = \sqrt{1 - (n_2/n_1)^2} = \sqrt{(n_1^2 - n_2^2) / n_1^2}$$

so

$$\sin \theta_a = n_1 \times \sqrt{(n_1^2 - n_2^2) / n_1^2} = \sqrt{(n_1^2 - n_2^2)}$$

The refractive index of the core has cancelled, which is a good sign — the answer depends on the *difference* between the two indices. Now put the numbers in:

$$\sin \theta_a = \sqrt{(1.5^2 - 1.2^2)} = \sqrt{(2.25 - 1.44)} = \sqrt{0.81} = 0.9$$

Answer: B, 0.9.

Why the other four are wrong.

- **0.8** is n_2/n_1 , the sine of the critical angle *inside* the fibre. It is an angle at the wall, not at the end face, and the two are different angles at different boundaries.
- **0.45** is half the correct value, as though only the core mattered and the cladding could be ignored.
- **1.0** would mean every ray entering the end face is guided, which would require the cladding to be absent altogether. With n_2 only slightly below n_1 the acceptance is indeed close to 1, but not equal to it.
- **1.2** quotes the cladding index as a sine. A sine of 1.2 does not exist, so the option is impossible on its face.

The wider point. The quantity $\sqrt{(n_1^2 - n_2^2)}$ is called the numerical aperture, and it is what optical engineers specify for a fibre, because it fixes how much light can be launched into it. Notice that a small difference between the two indices gives a small acceptance angle, which is why single-mode fibres are hard to couple light into.

Relevant topics: total internal reflection; Snell's law; critical angle; optical fibres.

The trap. Using $n_2/n_1 = 0.8$ as the answer. That is the sine of the critical angle at the core-cladding *wall*. The question asks for the acceptance angle at the *end face*, and converting between them needs the complementary-angle geometry and Snell's law.

Q22 · How long a kettle takes: electrical power meeting thermal capacity — **Answer E**

What is being tested. Whether you can chain two ideas together. Neither the electrical formula nor the thermal one is hard; the question is whether you connect them through energy.

Step 1 — how much energy the water needs.

$$Q = m c \Delta T$$

Step 2 — get the temperature rise right. This is where the marks are lost. The water starts at 20 °C and finishes at 100 °C, so the *rise* is

$$\Delta T = 100 - 20 = 80 \text{ K}$$

A temperature difference in kelvin has the same size as in degrees Celsius, so no conversion is needed — but the difference itself must be taken, not the final temperature. Substituting:

$$Q = 1.5 \times 4200 \times 80 = 504000 \text{ J}$$

Step 3 — the rate at which energy is supplied. A 2.0 kW kettle delivers energy at

$$P = 2.0 \text{ kW} = 2000 \text{ W} = 2000 \text{ J s}^{-1}$$

Step 4 — divide.

$$t = Q / P = 504000 / 2000 = 252 \text{ s}$$

Answer: E, 252 s.

Check it is sensible. Four minutes to boil a litre and a half is exactly what a kettle does. Any answer of a few seconds or a few hours should be rejected on sight, which is a useful first pass before any arithmetic.

Why the other four are wrong.

- **315 s** uses $\Delta T = 100 \text{ K}$, heating the water from 0°C . It is the single most common error here.
- **126 s** takes the kettle to be 4.0 kW .
- **504 s** takes the kettle to be 1.0 kW , so the answer is exactly twice the correct one — a neat way to spot it.
- **25 s** uses a specific heat capacity of 420 rather than 4200 .

The wider point. Questions that cross modules are usually joined by an energy chain, and the chain is worth writing out explicitly: *electrical energy in = thermal energy absorbed*. Once that is written, both sides are routine. The same structure appears in a pump raising water, a motor lifting a load, and a heater warming a room.

Relevant topics: electrical power; specific heat capacity; energy transfer; unit conversion.

The trap. Using a temperature rise of 100 K . The water starts at 20°C , so the rise is 80 K ; using 100 K inflates the answer by a quarter, which is exactly the difference between 252 s and 315 s .

Q23 · A charged capacitor connected to an uncharged one: where the charge goes — **Answer C**

What is being tested. Whether you use the right conservation law. The charge on the disconnected plates cannot change; the potential difference can.

Step 1 — the charge before connecting. The $2.0 \mu\text{F}$ capacitor is charged to 100 V , so

$$Q = C_1 V_1 = 2.0 \times 10^{-6} \times 100 = 200 \mu\text{C}$$

Step 2 — what is conserved. When the charged capacitor is disconnected from the supply and joined to the uncharged one, the two plates that are joined together form an isolated piece of conductor. No charge can leave it. So the total charge is still $200 \mu\text{C}$, now spread over both capacitors.

Step 3 — the combined capacitance. In parallel the capacitances add, because each capacitor has the same potential difference across it and their charges simply add:

$$C = C_1 + C_2 = 2.0 + 3.0 = 5.0 \mu\text{F}$$

Step 4 — the new potential difference.

$$V = Q / C = 200 \mu\text{C} / 5.0 \mu\text{F} = 40 \text{ V}$$

Answer: C, 40 V .

Check the direction of the change. The same charge now sits on a larger capacitance, so the potential difference must be *smaller* than 100 V . Any answer of 100 V or more is impossible, which removes one option immediately. And the answer must be more than zero, since charge did move. 40 V is the only option in range apart from 20 V and 50 V , and the arithmetic settles which.

Why the other four are wrong.

- **100 V** assumes no charge moves at all, which is what would happen if the second capacitor were not connected.
- **50 V** averages the two starting potentials. There is no reason for the answer to be an average of potentials; the weighting is by capacitance, and the correct weighting gives 40 V.
- **60 V** uses $C_2 V = 300 \mu\text{C}$ as the charge, i.e. the charge the *second* capacitor would hold at 100 V. The charge present is the one the first capacitor actually stored.
- **20 V** divides the 200 μC by 10 μF , adding a capacitance that is not there.

The wider point. Notice that this question cannot be answered by conservation of energy: joining two capacitors in this way always loses energy (in the resistance of the connecting wire, as heat and radiation). Charge is conserved, energy is not — the mirror image of the bullet-and-block question earlier in this section, where momentum was conserved and energy was not. Knowing which quantity survives a given process is most of the physics.

Relevant topics: capacitance; charge conservation; capacitors in parallel; energy dissipation.

The trap. Averaging the potentials to get 50 V. Charge is conserved, not potential: 200 μC spread over 5.0 μF gives 40 V. Note also that energy is *not* conserved here — the joining always loses some.

Q24 · A bubble rising in a lake: pressure at depth against volume — **Answer A**

What is being tested. Whether you include the atmosphere. The pressure at a depth is *not* just the water's contribution — it is the total pressure, and the atmosphere is part of it.

Step 1 — the pressure at the bottom. The pressure at a depth h in a liquid is the pressure at the surface plus the weight of the liquid above:

$$p_{\text{bottom}} = p_{\text{atm}} + \rho g h$$

Step 2 — evaluate the two parts.

$$\begin{aligned} \rho g h &= 1000 \times 10 \times 20 = 2.0 \times 10^5 \text{ Pa} \\ p_{\text{bottom}} &= 1.0 \times 10^5 + 2.0 \times 10^5 = 3.0 \times 10^5 \text{ Pa} \end{aligned}$$

So the pressure at the bottom is three atmospheres: one from the air and two from the 20 m of water. That "three" is the whole answer, and it is worth seeing it before doing any algebra.

Step 3 — the gas law at constant temperature. The bubble contains a fixed mass of gas (it does not gain or lose air) at a fixed temperature, so

$$p_1 V_1 = p_2 V_2$$

At the surface the pressure is the atmospheric pressure $1.0 \times 10^5 \text{ Pa}$.

Step 4 — solve for the new volume.

$$V_2 = V_1 \times p_1 / p_2 = 1.0 \times (3.0 \times 10^5) / (1.0 \times 10^5) = 3.0 \text{ cm}^3$$

Answer: A, 3.0 cm³.

Why the other four are wrong.

- **2.0 cm³** uses only the water's contribution, $2.0 \times 10^5 \text{ Pa}$, as the pressure at the bottom. The atmosphere is left out.
- **4.0 cm³** adds the atmosphere twice at the bottom, giving $4.0 \times 10^5 \text{ Pa}$.
- **1.5 cm³** would correspond to a depth of 5 m, where the water adds half an atmosphere.
- **1.0 cm³** leaves the volume unchanged, which would require the pressure to be the same at the bottom as at the surface. It is not — that is why the bubble grows as it rises.

A physical check. Divers are taught never to hold their breath while ascending, and this calculation is why: a lungful of air at 20 m expands to three times its volume by the time it reaches the surface. The same physics, with the same factor of three.

Relevant topics: pressure in a liquid; the gas laws at constant temperature; ratios; combining modules.

The trap. Using only pgh as the pressure at depth. The total pressure is $p_{\text{atm}} + pgh$; at 20 m the water contributes two atmospheres, so the bubble triples in volume, not doubles.

Q25 · What survives one and a half half-lives — **Answer D**

What is being tested. Whether you are comfortable with a fractional number of half-lives, and whether you can tell "remaining" from "decayed".

Step 1 — the decay law. After a time equal to n half-lives, the surviving fraction is

$$N / N_0 = (1/2)^n$$

This is just the statement that the quantity halves every time T passes. It is defined for any n , whole or not, because it is an exponential law.

Step 2 — put in $n = 1.5$. Split the power into a half and a square root, since $1.5 = 1 + 1/2$:

$$(1/2)^{3/2} = (1/2)^1 \times (1/2)^{1/2} = (1/2) \times 1/\sqrt{2} = 1 / (2\sqrt{2})$$

Step 3 — tidy the form. Rationalising the denominator gives an equivalent expression:

$$1 / (2\sqrt{2}) = \sqrt{2} / (2 \times 2) = \sqrt{2} / 4 \approx 0.35$$

So about 35 per cent of the original nuclei remain.

Answer: D, $1/(2\sqrt{2})$.

Check it against the whole-number cases. After one half-life, half remains — 0.5. After two, a quarter remains — 0.25. The answer for 1.5 half-lives must lie between those, and 0.35 does. Any option outside the range 0.25 to 0.5 is impossible without any algebra at all, which removes three of the five options immediately.

Why the other four are wrong.

- **$1/2$** is the fraction left after one half-life, i.e. after a shorter time than the question asks about.

- **$1/4$** is the fraction left after two half-lives. It rounds 1.5 up to 2, which is not allowed: the decay does not wait for a whole half-life to pass before doing anything.
- **$3/4$** adds the one-half-life and two-half-life fractions: $1/2 + 1/4 = 3/4$. Successive halvings *multiply*, they do not add. It is also worth noticing that $3/4$ would mean three quarters of the nuclei are still there after the source has halved twice over, which cannot be right.
- **$1/8$** is the fraction left after three half-lives, double the time asked about.

The wider point. Radioactive decay questions almost always have a quick range check available: the answer must lie between the values for the whole numbers of half-lives either side. Use that first, then do the algebra only on the options that survive.

Relevant topics: half-life; exponential decay; fractional powers; surds.

The trap. Adding the halvings instead of multiplying them, giving $1/2 + 1/4 = 3/4$. After one and a half half-lives about **0.35** of the nuclei remain, so any answer above $1/2$ is impossible.